

Principles of Atomic Force Microscopy

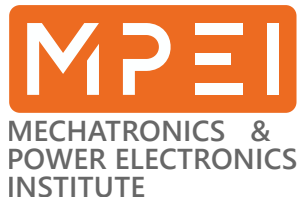
Mechatronic Systems Laboratory (369.002, LU, 2026S)
Update: 11.03.2026

Konstantin Waltenberger
konstantin.waltenberger@tuwien.ac.at
Florian Kreuzmair
florian.kreuzmair@tuwien.ac.at

Summer Semester, 2026
Update: 25.03.2026



TECHNISCHE
UNIVERSITÄT
WIEN



Contents

1	Introduction	3
2	AFM	3
2.1	System description	3
2.2	Tip-sample interaction and force curves	4
2.3	AFM modes	4
2.3.1	Contact mode	5
2.3.2	Tapping mode	5
3	Scaled-up AFM	5
3.1	System description of the scaled-up AFM	7
3.2	Differences between the scaled-up AFM and commercial AFMs	7
3.3	Direct demodulation of the cantilever oscillation amplitude	8
4	Exercise Instruction	9
4.1	Precautions	9
4.2	Preparation (Homework)	9
4.3	Experiments	9
4.4	Report	12

1 Introduction

An Atomic Force Microscope (AFM) is an instrument able to record images with atomic resolution, much higher than that of an optical microscope. In comparison with other types of high-resolution microscopes (e.g. Transmission Electron Microscopes (TEM)), AFMs have the advantage that they can be used with non-conductive samples both in ambient environments and in liquid. In this tutorial and laboratory exercise, the working principle of AFM is illustrated by means of a scaled-up prototype.

2 AFM

2.1 System description

The working principle of any AFM relies on the interaction between a microscopical cantilever beam with a sharp tip at its end (tip radius of $\sim 20\text{nm}$) and the surface of the sample to be measured. As the cantilever approaches the sample, it starts deflecting as a consequence of attractive and repulsive forces, such as the Van der Waals forces, the capillary forces, etc... By focusing a laser beam on the cantilever, which reflects onto a photo detector, this deflection can be measured and used in a feedback loop to regulate the vertical position of the cantilever. This ensures that the interaction force is kept constant and the cantilever and sample don't get destroyed. Such a regulation is usually achieved by means of a piezoelectric actuator, which expands itself proportional to the applied voltage (electric field) across its body. Via a calibration process, this voltage can then be used to reconstruct the quantitative topography of the sample.

Figure 1 shows the main components of an AFM in sample-scanning configuration. In this case, three actuators are employed: two to move the sample along the $x - y$ plane in a raster pattern, to allow the imaging of different points on the sample, and a third along the z axis for the topography measurement as described above. (In scanning-lever based AFMs the layout is flipped, and the cantilever is scanned, while the sample doesn't move.) Despite the high non-linearity inherent to piezoelectric materials (such as creep and hysteresis), most of the scanners used in AFM are based on piezoelectric actuators. This is due to their extremely high stiffness in the actuation, leading to a high imaging bandwidth.

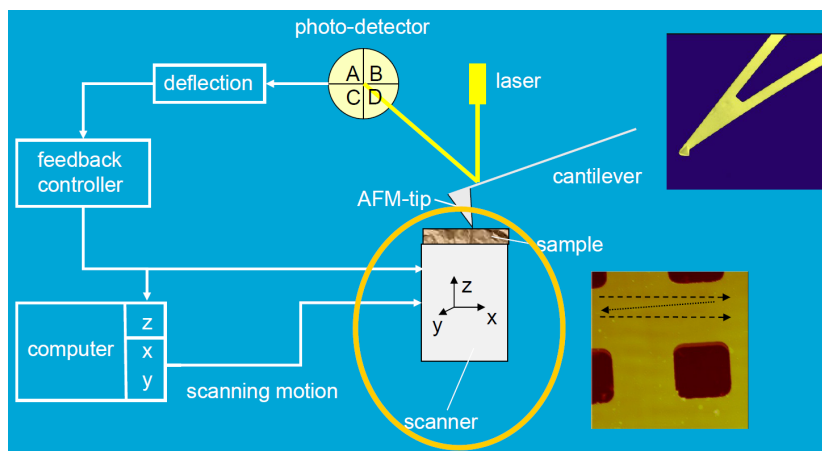


Figure 1: Main components and control loop of an AFM [1]

2.2 Tip-sample interaction and force curves

There are several forces acting on an AFM tip during scanning, such as the Van der Waals force, electrostatic force, friction and capillary force. These forces are responsible for the cantilever to deflect and are usually dependent on the distance between the sample and the cantilever tip. The relationship between the deflection and the distance is typically shown through *force curves*.

Fig. 2 shows a force curve when the tip approaches the sample (red solid line) and retracts from the sample (blue dotted line). When the tip is still far from the sample (**Point A** in Figure 2), there is no force acting on the tip, and the cantilever deflection is zero. As the tip gets sufficiently close to the sample (red solid line), attractive forces (e.g. Van der Waals force) become dominant and cause the tip to snap onto the sample with negative deflection (**Point B**). Further tip approach causes repulsive forces and changes the deflection driven by the stiffness between the cantilever and sample (**Point C**). When the tip retracts from the sample (blue dotted line), it typically shows a larger negative deflection than when it approaches (**Point D**), due to capillary (adhesive) forces. The adhesion usually originates from a thin liquid layer covering the sample at ambient conditions and can be changed by the structure of the molecules on its surface. For example, proteins (e.g. collagen fibrils) have multiple folding structures, resulting in multiple pull-offs caused by detachments within this folding structure. The differences in the force curve between approach and retraction provides valuable information of the sample properties such as elasticity, hardness, adhesion, surface charge density,... at the nanoscale. This analysis of the force curves is called force spectroscopy.

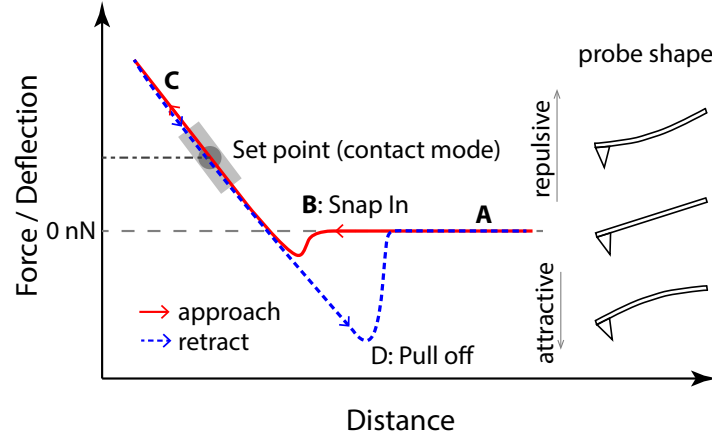


Figure 2: Simplified force curve with cantilever positions [2].

2.3 AFM modes

One of the major reason that makes AFM so popular in research is its huge variety of AFM modes, enabling various measurements of mechanical, electrical and even chemical properties of the sample on the nanoscale. In this section, the two most representative AFM modes (contact mode and tapping mode) for topography/height profile measurements are explained as basic AFM techniques. More details on AFM modes can be found in [2].

2.3.1 Contact mode

In contact mode, the tip is always in contact with the sample surface (**Point C** in Figure 2). In this configuration, the repulsive forces are measured by the cantilever deflection via Hooke's law. Typically, a feedback controller regulates the z-height, to keep the cantilever deflection and force constant, when scanning along the surface. This fixed force is called *set point* (shown as dark gray spot in Fig. 2) and is usually selected by user considering tip and sample properties (i.e. too high set-points cause high interaction forces, which might damage the sample or the tip). The feedback controller is designed to minimize the deflection error (= deflection - set-point). The *topography* of the sample can then be directly obtained by the controller output, with the deflection error being an indicator of *residual errors of the topography measurement*.

Contact mode provides a very simple configuration in terms of instrumentation and is suitable for high speed and high resolution imaging. As a drawback, the constant contact of the sharp cantilever with a certain force may degrade both the sample and the cantilever itself. This can be critical during measurements at ambient conditions as the capillary layer between the tip and the sample can cause large adhesive forces ($> \text{nN}$) during scanning, leading to image distortion or damage.

2.3.2 Tapping mode

In tapping mode (also called as intermittent contact mode) the cantilever oscillates above the sample, with the tip touching the surface periodically. This means that the tip swings between zero force regime **Point A** and repulsive regime **Point C** of Figure 2. The short-time tip-sample interaction results in a change of the amplitude and phase of this oscillation. Again, when scanning along a surface a feedback controller is used to keep the amplitude of the oscillation (with respect to a set-point) constant by controlling the z-height. The controller output is again used for the *topography* of the specimen and the amplitude-setpoint deviation can be interpreted as *residual errors of the measured topography*, similar to contact mode. On top of the topographic information, tapping mode also provides a *phase image*, which indicates an energy dissipation by the tip-sample interactions. Since this interaction differs, depending on materials, phase images can be used to identify the distribution of various materials over the surface.

Tapping mode is the most popular mode for imaging in air. That is because friction and capillary forces can be significantly reduced by the perpendicular tip movement to the sample surface. At ambient conditions, tapping mode can reduce the risks of sample damage and image degradation via short tip-sample interactions. Therefore tapping mode is widely used especially for imaging soft vulnerable samples such as biological tissue.

3 Scaled-up AFM

In this laboratory exercise, a scaled-up AFM is used to better understand the working principle of force curves, contact- and tapping mode.

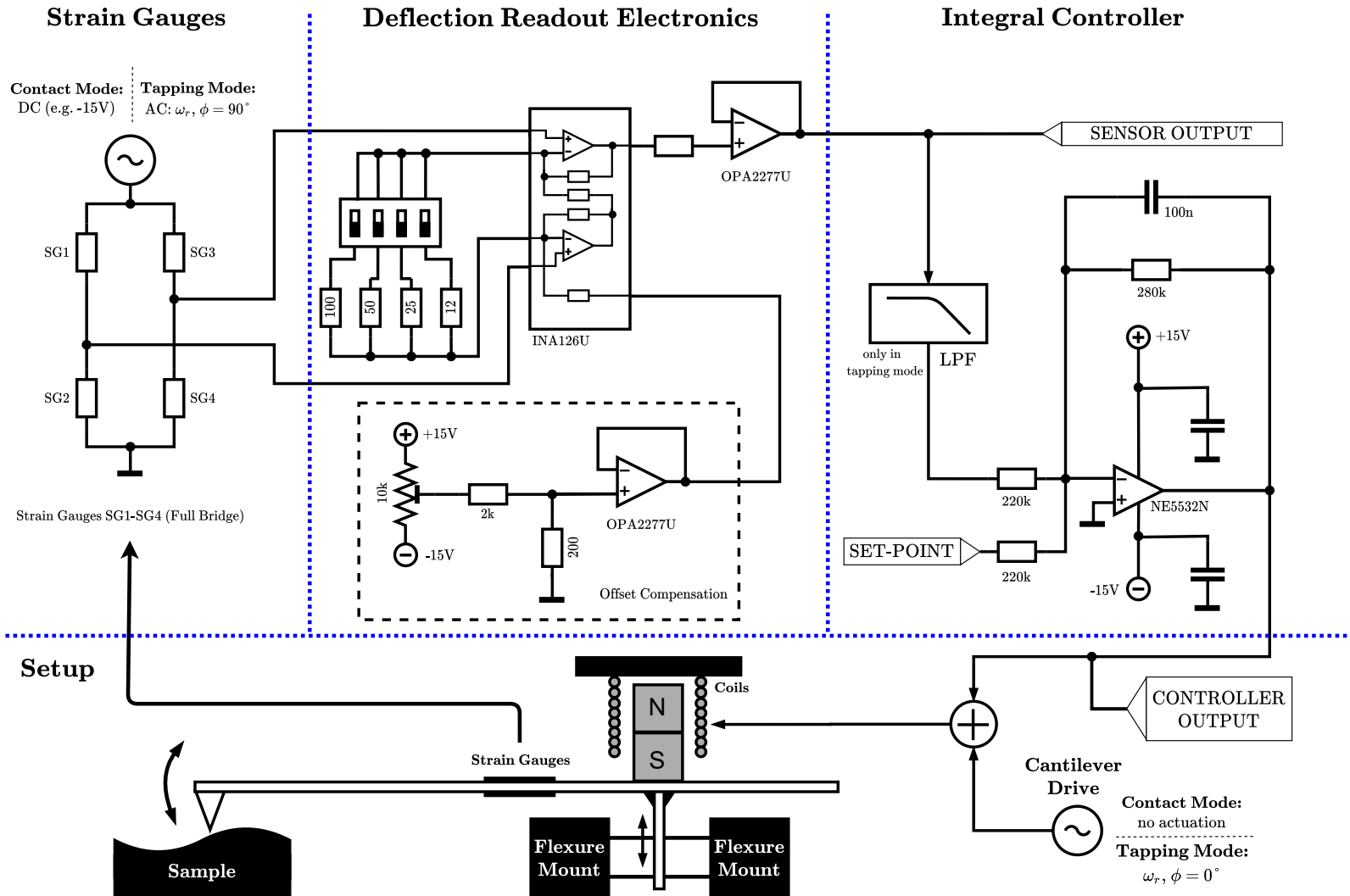


Figure 3: Schematics of the scaled-up AFM

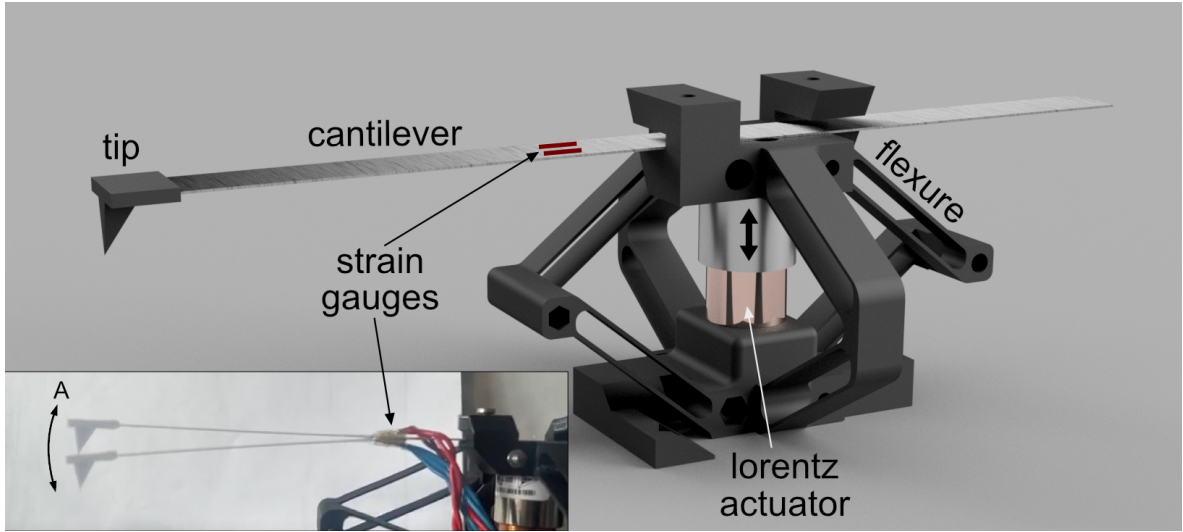


Figure 4: Rendering of the scaled-up AFM setup with an inset of the actual oscillating cantilever.

3.1 System description of the scaled-up AFM

Figure 4 shows a rendering of the scaled-up AFM used in this exercise. The setup uses an electromagnetic Lorentz actuator to actuate the cantilever vertically. Due to the low stiffness of such an actuator, a 3D-printed flexure is used that connects the cantilever to a fixed external frame to mimic the behavior of a piezoelectric actuator. To sense the cantilever deflection, four strain gauges are attached on the top and bottom side of the cantilever. Finally, a linear bearing stage is used to move the sample relative to the cantilever as a scanner. This stage has just one degree of freedom and is only for demonstrative purposes (real AFMs allow $x - y$ scanning).

The control blocks of the setup are shown in Figure 5. Figure 3 shows the entire circuit implementation of strain gauges, deflection readout electronics and an integral controller to generate the feedback loop for contact- and tapping mode AFM. During the course of the laboratory, the deflection readout electronics will be given and the students will have to build the controller. The strain gauges (SG_1 to SG_4) in the full bridge generate an output voltage proportional to the strain in the cantilever, caused by the deflection. As this voltage is extremely small, it is amplified by the instrumentation amplifier INA126 and treated as the deflection signal. The gain and offset of this amplification can be adjusted by the dip-switches and a variable resistor, respectively.

For the integral controller, the operational amplifier NE5532 adds the deflection signal to the set-point. The values of the electrical components and the circuit in Figure 3 are just for demonstrative purposes and should be modified. Analog circuits are discussed in more detail in the *Filter implementation* (FI) and *Sensor principles* (SP) tutorials.

3.2 Differences between the scaled-up AFM and commercial AFMs

To measure the cantilever deflection, three main methods are available in commercial AFMs. These are the optical interferometry method, the optical lever method and the piezoresistive

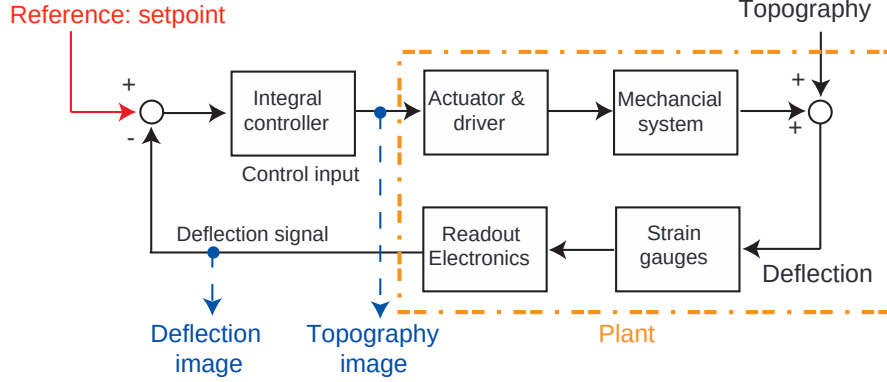


Figure 5: Control loop of an AFM in contact mode.

method. Among them, the optical lever method is the most popular due to its high sensitivity and simplicity to build. It relies on a laser module and a photo detector as illustrated in Figure 1. In such systems, the laser shines at the foremost point of the cantilever in order to maximize the deflection sensitivity. In the scaled-up AFM, the deflection is measured by measuring the strain of the cantilever by means of strain gauges. To maximize the sensitivity, the strain gauges have to be positioned where the strain is at maximum, i.e. at the base of the cantilever.

Sensing principle, sensor location and sensitivity are explained in more detail in the *Sensor Principles* (SP) and *Controllability and observability of dynamic systems* (CO) tutorials.

3.3 Direct demodulation of the cantilever oscillation amplitude

For closed-loop operation in tapping mode a measurement of the cantilever oscillation amplitude is necessary. In commercial AFM's this is usually achieved with the implementation of lock-in amplifiers, which multiply the deflection- with a sinusoidal reference-signal and subsequent low-pass filtering to suppress high frequency components. A more straightforward and practically simpler method to measure the oscillation amplitude is to demodulate directly within the strain-gauge bridge circuit [3]. This is achieved, by supplying the strain gauge bridge with an AC voltage, that has the same frequency as the cantilever actuation - see Fig. 6 and Eq. (1)-(3).

From calculating the output voltage U_{out} , while assuming the resistor values $R_1 - R_4$ and their changes due to stress ΔR are equal, follows:

$$U_{out} = U_S \cdot \left(\frac{R_2 + \Delta R}{R_1 + R_2} - \frac{R_4 - \Delta R}{R_3 + R_4} \right) \sim \frac{U_S}{2R} \cdot 2\Delta R. \quad (1)$$

Inserting for the supply voltage $U_S = U_0 \cdot \sin(\omega t + \phi_0)$ and the resulting resistor change $\Delta R = A \cdot K \cdot \sin(\omega t + \phi_c)$ due to actuation at ω (as indicated in Fig. 6), leads to:

$$U_{out} = \frac{U_0}{2R} \cdot A \cdot K \cdot [\cos(\phi_0 - \phi_c) - \cos(2\omega t + \phi_0 + \phi_c)]. \quad (2)$$

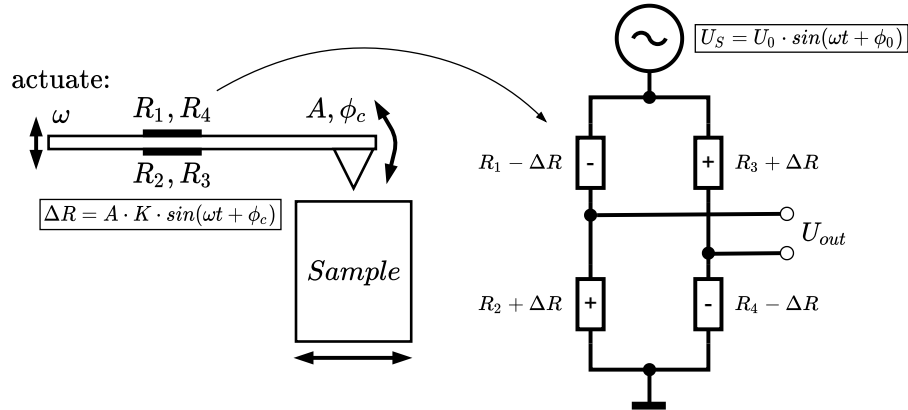


Figure 6: Direct cantilever amplitude demodulation within the strain gauge bridge circuit. (A = oscillation amplitude / ϕ_c = oscillation phase / K = strain gauge sensitivity)

Low-pass filtering the output voltage (as in Fig. 3) and maximizing the signal by setting ϕ_0 accordingly (i.e. $\phi_0 = \phi_c$), leads to the following final relation with the oscillation amplitude A :

$$U_{out} = \frac{U_0}{2R} \cdot A \cdot K \quad (3)$$

This output voltage is now directly proportional to the cantilever oscillation amplitude at the frequency ω and can be used for closed-loop feedback control.

4 Exercise Instruction

4.1 Precautions



Failing to respect the following warnings may cause serious damage to the equipment and/or to the setup!

- **The output current of the power supplies must not exceed $\pm 700 \text{ mA}$!**
Higher currents may cause the coils in the Lorentz actuator to burn if the feedback control becomes unstable. The voltage of the controller output should not exceed $\pm 400 \text{ mV}$ in this regard.
- **Do not bend the cantilever too much!**
Mechanical damage on the cantilever may result in wrong results .

4.2 Preparation (Homework)

There is no homework for this lab exercise.

4.3 Experiments

In the course of the following experiments, the working principle of contact mode and tapping mode of AFM are examined.

Calibration and Characterization of Sensor and Actuator

1. Adjust the offset of the deflection readout electronics (DRE) to obtain a 0 V deflection signal from the strain gauges, when cantilever is at resting position. **Do not change the potentiometer of the strain gauge!!! Do not change the gain of DRE!!! If you are not sure, please ask the tutors!!!** The deflection signal should be already visible (a few tens of mV) when bending the cantilever.
2. Characterize the Lorentz actuator by changing its supply voltage between 0.5 V and -0.5 V and measuring the tip height using a caliper as a function of the voltage itself. Discuss the results.
3. Identify the cantilevers resonance frequency, by applying a 10 Hz sine wave with a small amplitude ($< 100 mV_{pp}$) and sweeping the frequency up to 100 Hz. Measure the magnitude and phase response of the deflection as a function of drive frequency (Bode plot) using oscilloscope. Discuss the identified plant.
Note: Use the MATLAB script provided by the tutors and make sure that the tip does not touch the sample during this measurement. Use DC mode of oscilloscope.

Force Curve

4. **Do not connect the controller** and directly apply a triangular signal to the actuator (less than 1Hz). Measure the deflection/force to distance curve using the oscilloscope at the sticky and non-sticky area, and compare the results to Fig. 2. Discuss the results. (Use the round tip for this exercise)

Contact Mode Operation

5. Disconnect the Lorentz actuator from the voltage source.
6. Implement a controller. Study and discuss the provided controller (i.e. the circuit in Figure 3) on a breadboard. Give an estimate closed-loop control bandwidth. Note the parallel resistor in Fig.3 is removed in the implementation to achieve a pure integral controller. The capacitor can be discharged using the pushbutton. Engage the controller following the steps:
 - Approach the sample to the cantilever by using the z-stage until the bending of the cantilever is visible by strain gauge readout.
 - Check the strain gauge readout sign (+/-). When moving the z-stage (sample) up, the readout should decrease to guarantee stable feedback.
 - With the controller **unconnected** (open loop), adjust the setpoint of the controller using oscilloscope's wave generator to achieve approx. zero at the controller output.
 - Connect the controller to the Lorentz actuator, the system should be stabilized in a few seconds, with the controller output of approx. zero.
7. Calibrate the z-height readout of the system by measuring the calibration target on the sample.

8. Move the sample underneath the cantilever slowly and record the sensor and controller output signals. Try with different scanning velocities and setpoints (**moderate adjustments only**). How can each signal be interpreted? If the system becomes unstable during the adjustment, disconnect the Lorentz actuator and repeat Step 6.
9. Acquire a closed-loop system identification (Bode plot) using sine wave sweep (1-100 Hz), while keeping tip engaged to the sample. The amplitude should be kept below 100 mV_{pp} . Make sure the offset obtained in Step 6 is inserted correctly in the MATLAB script. Using AC mode of the oscilloscope for this identification. Discuss the results.

Tapping Mode Operation

10. **Disconnect the controller** and directly apply a small sine signal ($< 100\text{ mV}_{pp}$) at the resonance frequency of the cantilever to the actuator (use the signal generator and adder circuit provided by the tutor). **The activated mode shape should be the bending mode, avoid excitation of piston mode as much as possible.**

Approach the sample such that the tip touches the sample gently during the oscillation. The amplitude of the deflection signal should be reduced by approx. 20-30 % (in visual inspection). Move the sample underneath the cantilever (sticky and non-sticky area) and record the oscillation amplitude and the phase difference from the applied signal. Generate an image from each signal, i.e. amplitude and phase image. Discuss the influence of the sample adhesion properties. (Use the round tip for this exercise.)

11. **Direct demodulation (DD):** Use the synchronized second output of the signal generator to supply the strain gauge bridge at the same frequency as for the actuation ($\approx 20\text{ V}_{pp}$, do not mistake this with the actuation voltage!, start with phase offset $\phi_0 = 90^\circ$). Observe the sensor direct output on the oscilloscope. Turn off the actuator, and adjust offset in sensor readout, such that the DC component of sensor output is approximately zero.

Add low-pass filter (LP) to the sensor readout, observe the readout after the filter. Turn on the actuator again and **adjust the ϕ_0** in DD such that the sensor+LP reach approximately **positive maximum** (see Eq.2). Adjust the actuation amplitude of the actuator, check if the sensor+LP linearly scales with the actuation amplitude.

12. Choose any tip for this exercise, **please follow the following instructions STEP BY STEP to engage the controller!!!**
 - Approach the sample from the free-oscillation state, such that the oscillation amplitude reduces by approx. 20-30%.
 - Check the following **two conditions**: (1) the sensor+LP moves in negative direction, when the sample moves upwards. (2) the sensor readout is sensitive to sample height changes, e.g. $\geq 30\text{ mV}$ changes for 1mm height change.
 - If either condition is unsatisfied, please re-adjust the DD according to step 11.
 - Choose a sample height as 'height zero', and adjust the z-stage such that the oscillation amplitude is reduced by approx. 20-30% in total.
 - Adjust the offset in DRE to obtain an around zero sensor+LP output. This aligns the measurement zero with 'height zero'.

- With the controller **unconnected** (open loop), adjust the setpoint of the controller using oscilloscope's wave generator to achieve approx. zero at the controller output.
 - Connect the controller to the adder that drives the Lorentz actuator, the system should be stabilized in a few seconds, with the controller output of approx. zero.
13. Calibrate the sensor read out with the calibration target. Measure the ??? sample topography, and record controller output and sensor output. How are calibration and measurement compared to the same process in the contact mode? Discuss the process and results. If the system becomes unstable during the adjustment, disconnect the controller and repeat Step 12.
14. Change the sharp tip to the round tip, and repeat Step 12-13. (optional)

4.4 Report

Write a report, including all the experimental results. Also, add the following points:

- Plot the quantitatively measured topography of each section on the provided sample (use the reference measurement of the sample to compute the actual height from the voltage). Discuss the relationship between the AFM tip and the resolution of an AFM image.
- Analyzing the Bode plot, discuss the maximum control bandwidth achieved by the integral controller in contact mode.
- Discuss the tip-sample interaction in contact mode and tapping mode.

References

- [1] G. Schitter. Mechatronic systems: Atomic force microscope. *Course slides of Mechatronic Systems (376.050)*, 2013.
- [2] Peter Eaton and Paul West. *Atomic Force Microscopy*. Oxford university press, 2010.
- [3] Mathias Poik, Dominik Kohl, Mario Mayr, Christoph Kerschner, and Georg Schitter. A mechatronic lock-in amplifier: Integrating demodulation in sensor electronics for measuring mechanical oscillations. *IEEE Transactions on Instrumentation and Measurement*, 70:1–8, 2021.